Name: ______________________________________ **Class:** __________________

BLANTYRE SECONDARY SCHOOL CLUSTER

2019 MSCE MOCK EXAMINATION

CHEMISTRY

PAPER II (40 MARKS)

SUBJECT NUMBER: M038/II **Time Allowed: 2 hours**

Date: 28 March, 2019 **10:00 am onwards**

INSTRUCTIONS:

1. This paper contains **5 printed pages. Please check.**
2. Write your name on top of each page of the question paper.
3. Write your answers on the question paper.
4. This paper consists of **two** sections, **A** and **B.**
5. **Section A** consists of two descriptive questions on practical work to be answered in **1 hour**. Marks will be given for accurate and orderly presentation of facts supported by relevant diagrams.
6. In **section B** there are **two** practical questions to be answered in **1 hour**.
7. Marks for **section B** will be given for observation, accuracy and interpretation of results.
8. You should spend **30 minutes** on each question. The 30 minutes period for each question includes 3 minutes to tidy up apparatus and have it checked by the supervisor.

SECTION A (20 Marks)

1. In a laboratory, labels fell off from bottles containing an alkane, carboxylic acid, an alkanal and alkanone. Describe an experiment that could be done to identify the contents of the bottles.

(10 marks)

2. Describe an experiment that could be done to find the concentration of HCl using 0.1M NaCl by titration.

(10 marks)

SECTION B (20 Marks)

3. You are provided with 4 beakers, distilled water, measuring cylinder, sand paper and solutions of copper sulphate, zinc sulphate and magnesium sulphate. You are also provided with pieces of copper, zinc, iron and magnesium metals.

a. Pour about 2 cm^3 of copper sulphate solution into each of the four beakers.

b. Clean the copper, zinc, iron and magnesium metals using sand paper.

c. Put a piece of each metal into each of the four beakers containing copper sulphate solution.

d. Record the results in the table below, by indicating '**Reaction**' or '**No reaction**'.

e. Rinse the beakers with distilled water.

f. Repeat steps (a) to (f) using solutions of zinc sulphate, iron sulphate and magnesium sulphate respectively. **(9 marks)**

Table of Results

Metals / **Solutions**	**Copper**	**Zinc**	**Iron**	**Magnesium**
Copper Sulphate				
Zinc Sulphate				
Iron Sulphate				
Magnesium Sulphate				

g. Use the results to arrange the metals in order of increasing reactivity. **(1 mark)**

4. You are provided with 2 test tubes in a rack, a measuring cylinder, thermometer, spatula, tap water, ammonium chloride and sodium hydroxide pellets.

a. Pour 5cm^3 of water into each test tube.

b. Measure the initial temperature of water in each test tube and record the results in the appropriate spaces in the table.

c. Add half spatula of ammonium chloride (NH_4Cl) in one test tube and shake gently.

d. Measure the temperature of the ammonium chloride solution and record the results in the table.

e. Repeat steps **c** and **d** using pellets. **(4 marks)**

Table of Results

Solution	Initial temp. (˚)	Final temp. (˚)	Temp. change Final temp. – Initial temp. (˚)
Ammonium chloride (NH_4Cl)			
Sodium hydroxide (NaOH)			

f. State whether the change in each case is exothermic or endothermic

Ammonium chloride NH_4Cl) _______________________________________ **(1 mark)**

Sodium hydroxide (NaOH) ____ **(1 mark)**

g. Draw energy level diagrams to illustrate the dissolving of ammonium chloride and sodium hydroxide.

(3 marks)

h. State any one source of error in the experiment. **(1 mark)**

END OF QUESTION PAPER